

Past Teams Notes – C&C

Date:

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Researcher(s):

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Objective of research:

Look at past ROV teams and take notes on what we can learn from their projects

Notes:

San Diego Miramar College 2024 team (Sleepy Flshy):

- They used another kit (SeaMATE Barracuda ROV kit) for the electrical components
- For safety, they did all soldering with magnifying glasses, fume hoods, and lab doors open
- For safety, control board is equipped with fuses
- SID was made with AutoCAD Electrical

SFU Subvision 2024 team:

- Used Arduino Nano for microcontroller
- 12-volt power source
- Tried designing their own PCB but ended up burning it and to purchase an off-the-shelf PCB as it was cheaper

Desert Star Robotics 2024 team:

- Used fume fans when soldering and assembling electrical components
- Included fuses to avoid shorts, surges, and/or damaged electrical components
- 48-volt system
- Used components from MATE's Eagle Ray Kit
- Used Arduino Mega for microcontroller
- Electrical components were tested using a multimeter to see conductivity in the electrical works before use
- Had a pneumatics SID

UT Dallas 2024 Team (Hydromeda):

- Uses fuse and circuit breaker for safety
- Designed their own PCB
- PCB design had to account for the size of the ROV and allowed space

Triton Tech Robotics 2024 team:

- Uses 2 cameras
- Cameras stream in real-time
- One camera faced forward (helps with movement), other camera focuses on manipulator (helps with grabbing objects)
- 12-volt battery supply

Sources:

Pioneer 2024 technical documents: <https://drive.google.com/drive/folders/1J8gser7-oF44Zqw4PY7GGghtWxJiniDd>